# WEIQI CHI

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## EDUCATION

**THE UNIVERSITY OF TOKYO (QS #39, 2027)** TOKYO, JAPAN

Ph.D. student in Information Engineering Apr. 2024 – Mar.2027

** SPRING GX scholarship project student / Young Researcher's Encouragement Award, IEEE VTS 2024*

**TOKYO INSTITUTE OF TECHNOLOGY (QS #97, 2027)** TOKYO, JAPAN

Master's degree in Communication Engineering Oct. 2020 – Jun. 2022

** Supper Smart Society (SSS) scholarship project student*

## RESEARCH

**Learning-Based User Association & Mobility Management in mmWave V2X Networks** Current

*Doctoral Research*

Investigated contextual multi-armed bandit (CMAB) approaches for user association (UA) in mmWave vehicular networks under non-stationary channel conditions without CSI acquisition or offline training:

- Developed BAND algorithm by integrating change-point detection (CUSUM-CD) with blockage-aware reward estimation to overcome the detection failure of conventional bandits under non-stationary channel reward brought by vehicle mobility and transient blockages. *(IEEE GLOBECOM, 2026)*
- Proposed a novel trajectory-aligned knowledge (TAK) regions in a semi-distributed CMAB framework for cross-vehicle knowledge sharing and algorithm cold-start acceleration with rigorous regret-bound analysis. *(IEEE TVT)*
- Proposed DCC-SMAB algorithm using dynamic hierarchical context clustering to mitigate slow convergence in high-dimensional context spaces for learning efficiency promotion. *(IEEE VNC, 2026)*
- Developed simulation framework integrating SUMO traffic simulator and ray-tracing tools with real-world maps for realistic V2X communication data analysis *(IEICE RCS, 2025)*.
- Proposed a geometric-based dynamic V2I blockage modeling and detection method. *(IEEE VTC, 2024)*

**A Blockage-Aware mmWave V2X Connectivity Analysis based on MCM** Nov. 2024 – Feb. 2025

*Visiting Researcher at CARISSMA, Technische Hochschule Ingolstadt, Germany*

- Collaborated with Artery core development team to design and implement ETSI ITS-G5 standards-compliant blockage prediction system with cooperative awareness on Artery simulation platform.
- Developed Maneuver Coordination Message (MCM) based V2X network simulation evaluated with OMNeT++.

**Measurement and Prediction of Vegetation Attenuation on mmWave channel** Sep. 2020 – Jul. 2022

*Master Research & Joint Research with Rakuten Mobile, Inc.*

- Designed and conducted outdoor radio measurement campaigns for static shadowing effects from vegetation to 5G base stations, validated results through ray-tracing simulation with propagation models (DKED, ITU-R).
- Presented research findings at Rakuten Mobile group discussions and student conferences (Best Oral Presenter).

## FUNDINGS

**ACT-X, Japan Science and Technology Agency (JST)** Oct. 2026 – Mar. 2028

*Passed first-round screening and interview.*

**ACT-X** is one of the most prestigious grants available to PhD students in Japan, awarded by the country's leading national science funding body **JST** (comparable to NSF/DFG). Only 20 researchers within specified field were funded nationwide last year with direct funding of **¥6,000,000** (approx. **€33,000**) over 2 years.

## PUBLICATION

### Journal

[1] **Weiqi Chi**, B. Qian, H. Wu, Haibo Zhou, Manabu Tsukada, "AURA: An Automation-Level-Aware User Association and Resource Allocation Framework for AI-Native 6G V2X Networks," IEEE Communications Magazine. *(To be submit, **JCR Q1 ・IF 8.3**)*

[2] **Weiqi Chi**, B. Qian, H. Wu, D. Li, Haibo Zhou, Manabu Tsukada, "Toward Blockage-Resilient 6G-V2X Connectivity: Semi-Distributed Bandit with Dynamic Arm Set for mmWave HetNets," submitted to IEEE

Transactions on Vehicular Technology (**T-VT**), 2026. *(Under Review, **JCR Q1** ・**IF 7.5+**)*

### Internation conference

[1] H. Wu, B. Qian, H. Si, **Weiqi Chi**, E. Javanmardi, Haibo Zhou, Manabu Tsukada, " Sparsity and Evidential Uncertainty Enabled Communication and Sampling for Collaborative Semantic Occupancy Prediction," Proceedings of the AAAI Conference on Artificial Intelligence, 2027. *(Under Review, **CORE A***)*
[2] **Weiqi Chi**, Manabu Tsukada, "Blockage-Aware Non-stationary Dynamic Bandit for User Association in mmWave V2X Networks," IEEE Global Communications Conference (**GLOBECOM**) 2026. ***(CORE B)***
[3] **Weiqi Chi**, X. He, and Manabu Tsukada, "Fast-Adapting V2X Networks: A Dynamic Context Clustering Multi-Armed Bandits," IEEE Vehicular Networking Conference (**VNC**), Canada, Jun. 2026.
[4] **Weiqi Chi**, J. Nakazato, T. Murakami and Manabu Tsukada, "V2I Blockage Modeling and Performance Evaluation for Connected Autonomous Vehicle", The IEEE 99th Vehicular Technology Conference (**VTC2024-Spring**), Singapore, Jun. 2024. ***(VTS Flagship)***
[5] **Weiqi Chi**, N. Keerativoranan and Jun-ichi Takada, "Outdoor mmWave Band Tx Localization Using Scattering Path under Spectrum Sharing Scenario," 13th Asia-Oceania Top University League on Engineering (**AOTULE**) Conference, Daejeon, Korea, Nov. 2021.

### Domestic conference

[1] **Weiqi Chi**, N. Keerativoranan, Jun-ichi Takada, "Outdoor Measurement of Millimeter-Wave Propagating Through Vegetation at 28GHz Band," MISW Conference, Tokyo, Japan, Aug. 2022. *(Chairperson)*
[2] **Weiqi Chi**, J. Nakazato, T. Murakami, Manabu Tsukada, "ミリ波帯における遮蔽対応型 V2I 通信と解決策について", in IEICE 無線通信システム研究会, Kyoto Institute of Technology, Mar. 2025.

### Poster

[1] **Weiqi Chi** and M. Tsukada, "Bridging the Gap: A Realistic V2X Simulation for End-to-End Autonomous Driving," in Cooperative Interactive Vehicles (CIV) Summer School 2026, Japan, Tokyo, 2026.
[2] **Weiqi Chi** and M. Tsukada, "Blockage-aware Multi-armed Bandits for Proactive Handover in mmWave Vehicular Networks," in Cooperative Interactive Vehicles (CIV) Summer School 2025, Geneva, Switzerland, 2025.

## OTHER EXPERIENCE

### Reviewer

*IEEE Transactions on Cognitive Communications and Networking (TCCN)*

**IEEE INFOCOM 2026, Tokyo, Japan** May. 2026

*Student Volunteer* | *Served as cameraman and logistics support staff for 900+ attendees from 40 countries*

**Visban Inc., Tokyo, Japan** Jun. 2026 – Current

*Intern* | *mmWave repeater simulation engineer* | Sionna Raytracing, Network Controlled Repeater

**Tier IV Inc., Tokyo, Japan** Apr. 2026 – Sep. 2026

*Intern* | *Autoware V2X engineer* | Autoware, Veins, QuaDRiGa

**Ehime Prefectural Industrial Technology Institute, Japan** Aug. 2025

*Intern* | *AR/VR Application Development* | MRTK, Unity, C#, HoloLens2

**General Test Systems Inc., Shenzhen, China** Feb. 2023 – Dec. 2023

*Intern* | *Application engineer* | OTA testing, Anechoic chamber, VNA, Area Tester

**Rakuten Mobile Inc. R&D group, drone application development, Japan** Sep. 2022

*Intern* | *Drone application development* | DJI drones operation, SDK, Android platform

## SKILL

- **Programming skills:** C, C++, Python, MATLAB, NED language
- **ML Frameworks & Data processing:** Scikit-learn, NumPy, Pandas
- **Development & Tools:** Git, Docker, LaTeX, Linux/Ubuntu
- **Hardware:** Vector Network Analyzer (VNA), Spectrum Analyzer, OTA testing
- **Simulation & Modeling:** OMNeT++, SUMO, Artery, Veins, Wireless InSite, Sionna, Unity, Blender
- **Language:** Chinese (Native), English (Native level) , Japanese (JLPT N1) , German (Beginner)
- **Interests:** Piano, Music, Swimming, Boxing, Bouldering